

Highlights

Logarithmic Weighted Cross-Entropy: A Stable and Parameter-Efficient Re-weighting Loss for Class-Imbalanced Classification

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- A logarithmic re-weighting loss that bounds the weight explosion of weighted cross-entropy.
- A single-parameter generalization tuned automatically on a small proxy subset.
- Evaluation spans image, tabular, and network-intrusion domains of varying imbalance.
- The proposed losses improve as imbalance grows and never collapse on any dataset.

Logarithmic Weighted Cross-Entropy: A Stable and Parameter-Efficient Re-weighting Loss for Class-Imbalanced Classification

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ABSTRACT

Class imbalance is a pervasive obstacle in machine learning, arising whenever a few majority categories dominate the training data while critically important minority categories remain severely underrepresented. Among algorithm-level remedies, Weighted Cross-Entropy (WCE) is the most direct: it scales the loss of each class by the inverse of its frequency. However, these inverse-frequency weights grow without bound as the imbalance becomes more severe, producing unstable optimization. More elaborate alternatives such as Class-Balanced (CB) Loss and Focal Loss introduce additional hyperparameters and, in our experiments, behave inconsistently across datasets—occasionally falling below ordinary Cross-Entropy (CE). We propose Logarithmic Weighted Cross-Entropy (LWCE), which replaces the inverse-frequency weight with a logarithmically compressed counterpart that keeps a meaningful boost for rare classes while bounding weight growth, and which introduces no additional hyperparameters. We further generalize it to Power-Log Weighted Cross-Entropy (PLWCE) through a single exponent that adjusts the focusing strength and is selected automatically on a small proxy subset. To assess generality, we evaluate the proposed losses across three distinct application domains that together span a wide range of imbalance severity: long-tailed image classification, tabular classification benchmarks, and network intrusion detection. Across these domains, LWCE and PLWCE improve over Cross-Entropy as the imbalance grows and, unlike the competing re-weighting losses, never collapse to substantially worse performance on any dataset. These results indicate that logarithmic weight compression offers a simple, stable, and parameter-efficient alternative to existing re-weighting strategies for class-imbalanced classification.

1. Introduction

Across a wide range of real-world classification tasks—including medical diagnosis, financial fraud detection, network intrusion detection, and visual recognition—the categories of interest are rarely observed in equal proportions. Instead, a small number of majority categories concentrate the overwhelming majority of the available samples, while many minority categories are represented by only a handful of examples (He and Garcia, 2009; Johnson and Khoshgoftaar, 2019). This long-tailed structure is not an artifact of poor data collection but an intrinsic property of the underlying phenomena: healthy patients vastly outnumber those with a rare disease, legitimate transactions dwarf fraudulent ones, and benign network traffic far exceeds malicious traffic. Because the minority categories are often precisely the ones that carry the greatest practical cost when missed, building classifiers that remain reliable under imbalance is a problem of broad and lasting importance.

The central difficulty is that the standard training objective is itself biased toward the majority. When a model is trained with ordinary Cross-Entropy (CE) loss, the aggregate gradient signal is dominated by the numerically abundant classes, so the optimizer can drive the overall loss down while largely ignoring rare categories. The resulting classifier attains a high overall accuracy yet performs poorly on exactly the minority classes that matter most—a failure mode commonly described as majority bias. Standard accuracy therefore masks the problem rather than revealing it, and a method that addresses imbalance must improve minority-class performance without sacrificing the majority.

Existing remedies fall broadly into two families. Resampling methods reshape the training distribution directly: oversampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) synthesize new minority instances (Chawla, Bowyer, Hall and Kegelmeyer, 2002), while undersampling discards majority examples to restore

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balance. Both directions carry well-known risks—oversampling can introduce synthetic artifacts and encourage overfitting, whereas undersampling discards potentially informative majority data (Drummond and Holte, 2003). Algorithm-level methods instead modify the training objective. These include margin-based losses that enlarge the decision boundary around minority classes (Cao, Wei, Gaidon, Aréchiga and Ma, 2019), logit-adjustment techniques that calibrate class priors, and loss re-weighting methods that scale the per-class loss by class-dependent weights. Among these, re-weighting is especially attractive because it is architecture-agnostic, modality-independent, and requires only a minimal change to the loss function; this paper focuses on the re-weighting family.

Despite its appeal, re-weighting remains unsatisfying in practice. Weighted Cross-Entropy (WCE), the most direct re-weighting scheme, assigns each class a weight inversely proportional to its frequency; as the imbalance grows more severe these weights grow without bound, producing extreme gradients that destabilize training. More sophisticated alternatives were designed to soften this behavior but bring their own drawbacks. Focal Loss reduces the influence of easy, typically majority, examples through a tunable focusing term, but it operates on per-sample confidence rather than class frequency and therefore behaves inconsistently across imbalance levels. Class-Balanced (CB) Loss re-weights by an effective number of samples governed by an additional hyperparameter, yet that hyperparameter is sensitive and, in our experiments, CB Loss degrades below plain Cross-Entropy under severe imbalance. In short, the field still lacks a re-weighting loss that is simultaneously simple, hyperparameter-light, numerically stable, and dependable across both the level of imbalance and the application domain.

To address this gap, we propose Logarithmic Weighted Cross-Entropy (LWCE). The key idea is to replace the inverse-frequency weight of WCE with a logarithmically compressed counterpart. The logarithm fundamentally bounds the growth of the weights, so the explosive behavior of WCE is removed while a meaningful emphasis on rare classes is preserved; crucially, the resulting loss introduces no additional hyperparameters and can be used as a drop-in replacement for Cross-Entropy. We further generalize LWCE to Power-Log Weighted Cross-Entropy (PLWCE) by introducing a single exponent that continuously controls the focusing strength, ranging from near-uniform weighting to an aggressive emphasis on the tail. This exponent is the only free parameter, and it is selected automatically through a lightweight grid search on a small proxy subset, adding negligible computational overhead.

We assess the proposed losses through a deliberately broad empirical study rather than a single benchmark. To test whether the benefit of logarithmic compression generalizes, we evaluate across three application domains that together cover a wide spectrum of imbalance severity: long-tailed image classification on standard vision benchmarks, a collection of tabular classification datasets drawn from public repositories, and several network intrusion detection datasets in which rare attack types are extremely scarce. In every domain we compare the same set of re-weighting losses under a controlled protocol with repeated runs, and we report metrics that explicitly reflect minority-class performance rather than overall accuracy alone. This cross-domain design lets us examine not only whether the proposed losses help, but how their advantage changes as imbalance becomes more extreme.

The main contributions of this paper are as follows.

1. We identify and characterize the weight-explosion failure mode of Weighted Cross-Entropy and motivate logarithmic compression as a principled remedy.
2. We propose Logarithmic Weighted Cross-Entropy, a parameter-free drop-in replacement for Weighted Cross-Entropy, together with its single-parameter generalization, Power-Log Weighted Cross-Entropy, whose only exponent is tuned automatically on a small proxy subset.
3. We provide a controlled cross-domain evaluation spanning image, tabular, and network intrusion detection data, showing that the proposed losses improve over Cross-Entropy as imbalance grows and, unlike Class-Balanced Loss and Focal Loss, never collapse to substantially worse performance on any dataset.

The remainder of this paper is organized as follows. Section 2 reviews related work on learning from imbalanced and long-tailed data. Section 3 defines the proposed losses and analyzes their properties. Section 4 describes the experimental setup across the three domains and presents the results. Section 5 concludes and discusses limitations and future work.

2. Related Work

3. Proposed Method

4. Experiments

5. Conclusion

CRedit authorship contribution statement

Seung Ho Go: Conceptualization, Methodology, Software, Writing – original draft. **Hyunseok Lee:** Methodology, Validation, Writing – review & editing. **Jin Hee Yoon:** Supervision, Conceptualization, Writing – review & editing.

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